EERoom

Browser-Based Electrical Engineering Simulation Platform

Final Requirements Specification, Revision 1.3

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Course Number	CST-452 Senior Project II
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Revision	1.3 (final, supersedes 1.2 from Milestone 5)
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Abstract

This is the final Requirements Specification for EERoom, describing the requirements as delivered in the released application at www.eeroom.io. No requirement changed after the third code iteration; Revision 1.3 confirms the final status of every requirement and use case for Milestone 6. Revision 1.1 recorded every use case, user story, technology, and tool taken out of scope at Milestone 4, with the justification for each. Revision 1.2 closes the two items that were deferred: the password reset form (SC-07), which completes FR-04, and guest access to the tools (SC-06). It also records one tool change, Vitest instead of Jest for the server API tests (SC-08), and adds NFR-15 for text contrast. The removals from Revision 1.1 still need approval from the mentor and the instructor on the signoff sheet.

History and Signoff Sheet

Date	Author	Revision	Notes
CST-451	Aiden Vargas	1.0	Original Requirements Specification (Milestone 2)
10/11/2026	Aiden Vargas	1.1	Scope adjustments after the second code iteration (Milestone 4)
10/25/2026	Aiden Vargas	1.2	Third code iteration: FR-04 complete, guest access added, Vitest for API tests, NFR-15 added (Milestone 5)
11/04/2026	Aiden Vargas	1.3	Final release: status confirmed for every requirement (Milestone 6)

Scope Change Approval

Change	Summary	Mentor	Instructor
SC-01	Remove the DC circuit simulator	☐	☐
SC-02	Remove Google OAuth login	☐	☐
SC-03	Remove Stripe, the Pro tier, the save limit, and PDF export	☐	☐
SC-04	Grid placement in SVG instead of react-konva drag and drop	☐	☐
SC-05	Ideal-versus-sampled overlay instead of a three-signal overlay	☐	☐
SC-06	Defer guest access to the tools (resolved at Milestone 5)	☐	☐
SC-07	Defer the password reset form (resolved at Milestone 5)	☐	☐
SC-08	Vitest with Supertest instead of Jest with Supertest for API tests	☐	☐

Submitted by	Aiden Vargas, 11/04/2026
Instructor signature and date	
Mentor signature and date	

1. Reason for This Revision

After the first code iteration in CST-451, the full plan was larger than one developer could build and test well in the remaining weeks. The highest-risk items were removed so the remaining features could be finished, deployed, and tested properly. The three simulation tools that remain still cover calculation, signals, and digital logic, which is the core purpose of EERoom.

Every removal below follows the same rule: keep what proves the platform works for students, and drop what adds risk without adding learning value for this MVP.

2. Items Taken Out of Scope

2.1 Use Cases and Functional Requirements

Change	Requirements	What Changes	Justification
SC-01	FR-06 to FR-11	Remove the DC circuit simulator	Solving a conductance matrix for any user-built circuit was the riskiest feature in the plan. It could not be finished and tested by one developer in the time left.
SC-02	FR-02	Remove Google OAuth login	Email and password login through Supabase Auth already meets the authentication need. OAuth adds a third-party integration with no added value here.
SC-03	FR-32 to FR-36	Remove the save limit, Pro tier, and Stripe billing	Payments are not needed to show the platform works. Removing them also removes a payment security risk. All users get unlimited saves.
SC-04	FR-17	Place gates on a grid instead of free drag and drop	Grid cells give each wire a clear endpoint and make the tool easier to use and demonstrate. Wires can still be dragged between pins.
SC-05	FR-25	Overlay the ideal and sampled signal instead of three signals	Showing what a chosen sample rate records teaches aliasing, a key signals concept, and keeps saved data simple.

2.2 User Stories

ID	User	User Story	Status	Note
US-02	Student	Create an account with Google login	Changed	SC-02. Email and password instead
US-03	Student	Drag resistors and voltage sources onto a circuit canvas	Removed	SC-01
US-09	Pro user	Save unlimited projects	Changed	SC-03. Unlimited for every user
US-10	Pro user	Export a simulation as a PDF	Removed	SC-03
US-11	Student	See an upgrade prompt at the save limit	Removed	SC-03
US-12	Pro user	Cancel a subscription from settings	Removed	SC-03

2.3 Technologies and Tools

Technology or Tool	Original Purpose	Status	Justification
Konva.js and react-konva	Canvas drag and drop for circuit and logic tools	Removed	Circuit simulator removed; logic tool draws in inline SVG on a grid (SC-01, SC-04)
math.js	Formula evaluation and matrix solving	Removed	Matrix solving was only needed for the circuit simulator; Ohm's Law uses plain arithmetic

Technology or Tool	Original Purpose	Status	Justification
Google OAuth 2.0	Social login	Removed	Email and password through Supabase Auth meets the requirement (SC-02)
Stripe	Subscription billing	Removed	Payments removed from the academic MVP (SC-03)
express-rate-limit (MIT)	Rate limiting on public auth routes	Added	Implements the Security Design's limit of 10 attempts per 15 minutes
Vitest (MIT)	Automated testing	Added	82 frontend unit tests, and the runner for the 37 server API tests (SC-08: used in place of Jest so one runner covers both folders)
Supertest (MIT)	HTTP testing of the Express API	Added	Sends requests to the app in the server tests without opening a port
Jest	Server API testing (planned in the Milestone 4 backlog)	Removed	Replaced by Vitest (SC-08); Vitest already runs the frontend tests and supports the server's ES modules without extra setup
Custom domain eeroom.io	Production address	Added	Connected to the Vercel deployment

2.4 Deferred Items Resolved at Milestone 5

- SC-06: Visitors can now use all three tools without an account. Saving still requires one, and a guest's work is saved to the new account automatically after they sign up and log in.
- SC-07: The Forgot Password and Reset Password pages are built, completing FR-04. The reset request gives the same answer for every email so it cannot reveal which accounts exist.

3. Functional Requirements Status

All 39 functional requirements, with their final status: 22 implemented, 2 changed, 0 partial, 12 removed, and 3 added.

ID	Requirement	Status	Note
FR-01	Register a new account with email and password	Implemented	
FR-02	Log in using Google OAuth	Removed	SC-02. Email and password login only
FR-03	Log out from any page	Implemented	
FR-04	Reset password through an email link	Implemented	Completed in the third iteration (SC-07 resolved)
FR-05	Show an error message for invalid login credentials	Implemented	
FR-06	Drag and drop circuit components onto a canvas	Removed	SC-01. Circuit simulator removed
FR-07	Connect circuit components by drawing wires	Removed	SC-01
FR-08	Calculate voltage and current at each node using DC analysis	Removed	SC-01
FR-09	Update circuit results in real time	Removed	SC-01
FR-10	Delete circuit components or clear the canvas	Removed	SC-01
FR-11	Show editable component values on the circuit	Removed	SC-01
FR-12	Input any two of Voltage, Current, and Resistance	Implemented	
FR-13	Calculate the third value with Ohm's Law	Implemented	
FR-14	Calculate and display power	Implemented	
FR-15	Update calculated values instantly while typing	Implemented	
FR-16	Display units next to all values	Implemented	Also guards against division by zero
FR-17	Provide AND, OR, NOT, NAND, NOR, and XOR gates	Changed	SC-04. Click-to-place grid; XNOR added
FR-18	Connect gate inputs and outputs with wires	Implemented	By click or drag; wires can be removed
FR-19	Evaluate the circuit and show output states	Implemented	
FR-20	Generate a truth table for the circuit	Implemented	
FR-21	Toggle input values between 0 and 1	Implemented	
FR-22	Select sine, square, sawtooth, or triangle signals	Implemented	
FR-23	Configure amplitude, frequency, and phase	Implemented	DC offset added
FR-24	Render the signal on an oscilloscope-style display	Implemented	

ID	Requirement	Status	Note
FR-25	Overlay up to three signals for comparison	Changed	SC-05. Ideal and sampled traces overlaid
FR-26	Label axes with time and amplitude units	Implemented	
FR-27	Save the current simulation as a named project	Implemented	
FR-28	Load saved projects from a dashboard	Implemented	
FR-29	Delete saved projects	Implemented	Inline confirmation added
FR-30	Generate a unique shareable URL	Implemented	
FR-31	View a shared project without logging in	Implemented	
FR-32	Limit free accounts to 3 saved projects	Removed	SC-03. Unlimited saves for all users
FR-33	Show a Pro upgrade prompt at the save limit	Removed	SC-03
FR-34	Process $9 per month Pro payments through Stripe	Removed	SC-03
FR-35	Grant Pro access after payment	Removed	SC-03
FR-36	Let Pro users cancel their subscription	Removed	SC-03
FR-37	Set a sample rate and warn when the signal is aliased	Added	Added in the second iteration
FR-38	Derive a boolean expression for each logic output	Added	Added in the second iteration
FR-39	Flag unconnected gate pins and refuse feedback loops	Added	Added in the second iteration

4. User Story Status

ID	User	User Story	Status	Note
US-01	Guest	View a shared simulation link without an account	Implemented	Guests can also use every tool without an account (SC-06)
US-02	Student	Create an account with Google login	Changed	SC-02. Email and password instead
US-03	Student	Drag resistors and voltage sources onto a circuit canvas	Removed	SC-01
US-04	Student	Use the Ohm's Law calculator for homework	Implemented	
US-05	Student	Build a logic circuit and get a truth table	Implemented	
US-06	Student	Visualize a sine wave at 60 Hz	Implemented	
US-07	Student	Save a project and share it with a professor	Implemented	
US-08	Student	Reopen saved projects from a dashboard	Implemented	
US-09	Pro user	Save unlimited projects	Changed	SC-03. Unlimited for every user
US-10	Pro user	Export a simulation as a PDF	Removed	SC-03
US-11	Student	See an upgrade prompt at the save limit	Removed	SC-03
US-12	Pro user	Cancel a subscription from settings	Removed	SC-03

5. Non-Functional Requirements

Non-functional requirements NFR-01 to NFR-12 from Revision 1.0 are retained without change. NFR-13 and NFR-14 were added in Revision 1.1, and NFR-15 in Revision 1.2:

ID	Category	Requirement
NFR-13	Security	Public authentication routes shall allow no more than 10 requests per 15 minutes from one address.
NFR-14	Maintainability	Core calculation logic and the server API shall be covered by automated tests that run with one command in each folder.
NFR-15	Accessibility	Body text shall have a contrast ratio of at least 4.5:1 against its background, and interactive targets shall be at least 24 px across (WCAG 2.2 AA).